



Association-Based Spectral Clustering for Mixed Data

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Roadmap: Distance-based learning with mixed data

Line of work: increasing **awareness** in distance construction

Scale/type-aware

Additivity, commensurability, and bias in mixed data

Association-aware

Redundancy, correlations, and categorical associations

Interaction-aware

Continuous–categorical relationships and local neighbourhood structure

Response-aware

Supervised neighbourhoods and prediction-oriented distances



Building distance-based pipelines: **manydist**¹

Distance construction, diagnostics, and learning workflows

Today's focus

Line of work: increasing **awareness** in distance construction

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Supervised neighbourhoods and prediction-oriented distances

↪ Carlo's talk



Building distance-based pipelines: **manydist**¹

Distance construction, diagnostics, and learning workflows

↪ Angelos' talk

setup

Learning from distances

Many learning methods can operate directly on a dissimilarity matrix.

Dimension reduction

Multidimensional scaling (MDS)¹

Clustering methods

Hierarchical clustering (HC), PAM²,
and **spectral clustering**³

Nearest-neighbour prediction

Nearest-neighbour classification
and nearest-neighbour averaging
for regression

choosing a different **distance** can lead to a different downstream analysis result

Mixed-data setup

a mixed data set

- I observations described by Q variables, Q_n numerical and Q_c categorical
- the $I \times Q$ data matrix $\mathbf{X} = [\mathbf{X}_n, \mathbf{X}_c]$ is column-wise partitioned

A formulation for mixed distance between observations i and ℓ :

$$\begin{aligned} d(\mathbf{x}_i, \mathbf{x}_\ell) &= \sum_{j_n=1}^{Q_n} d_{j_n}(\mathbf{x}_i^n, \mathbf{x}_\ell^n) + \sum_{j_c=1}^{Q_c} d_{j_c}(\mathbf{x}_i^c, \mathbf{x}_\ell^c) = \\ &= \sum_{j_n=1}^{Q_n} w_{j_n} \delta_{j_n}^n(\mathbf{x}_i^n, \mathbf{x}_\ell^n) + \sum_{j_c=1}^{Q_c} w_{j_c} \delta_{j_c}^c(\mathbf{x}_i^c, \mathbf{x}_\ell^c) \end{aligned}$$

numeric case

- $\delta_{j_n}^n$ is a function quantifying the dissimilarity between observations on the j_n —th numerical variable
- w_{j_n} is a weight for the j_n —th variable.

categorical case

dissimilarity between the categories chosen by subjects i and ℓ for categorical variable j_c

- w_{j_c} is a weight for the j_c —th variable

Independence-based distances

Independence-based pairwise distance

No inter-variable relations are considered.

- in the continuous case: **Euclidean** or **Manhattan** distances
- in the categorical case: **Hamming** / matching distance, among many others
- in the mixed-data case: **Gower** dissimilarity index

Beyond commensurability

commensurability makes variable contributions comparable across **scales** and data **types**¹.

- If variables are correlated or associated, the same information may contribute repeatedly to the distance:
redundancy

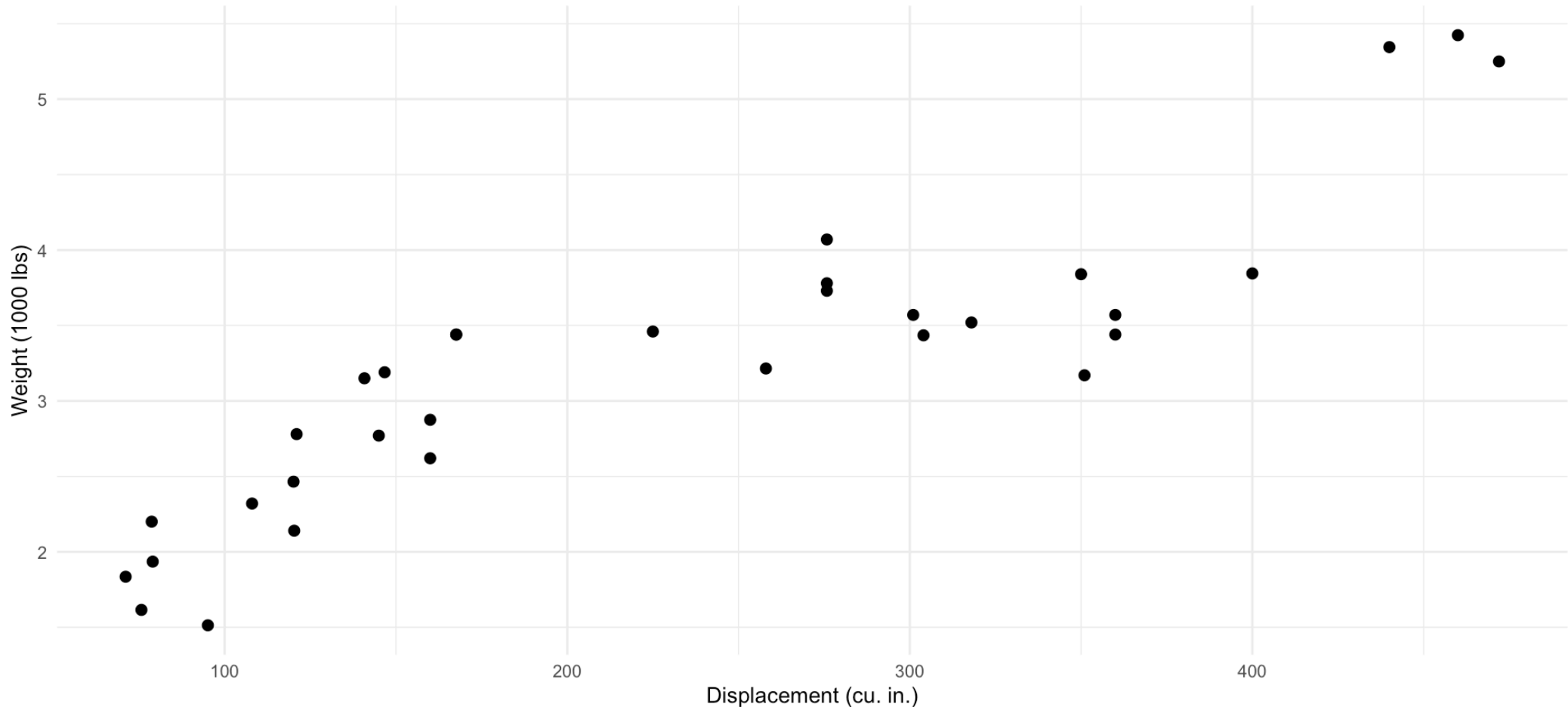
Association-aware distances

by variable differences: independence-based

- When variables are correlated or associated, shared information is effectively counted multiple times
- inflated dissimilarities may cause potential distortions in downstream unsupervised learning tasks.

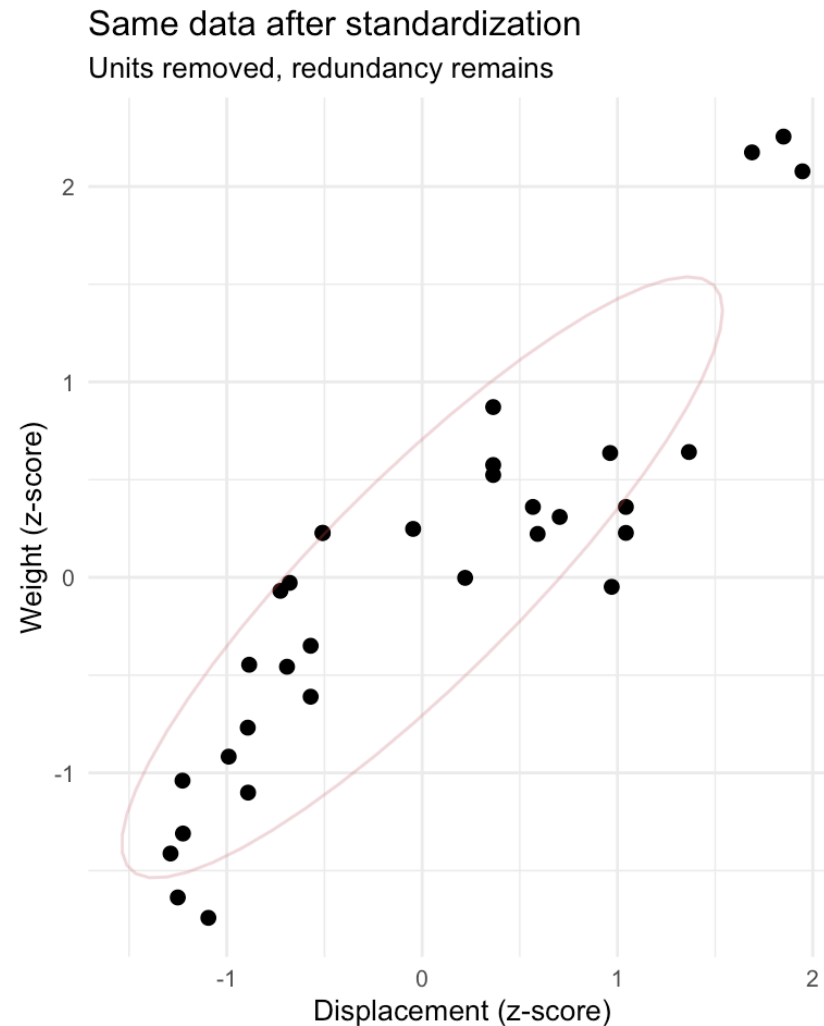
Redundant continuous variables: displacement and weight

Larger cars tend to have both higher displacement and weight



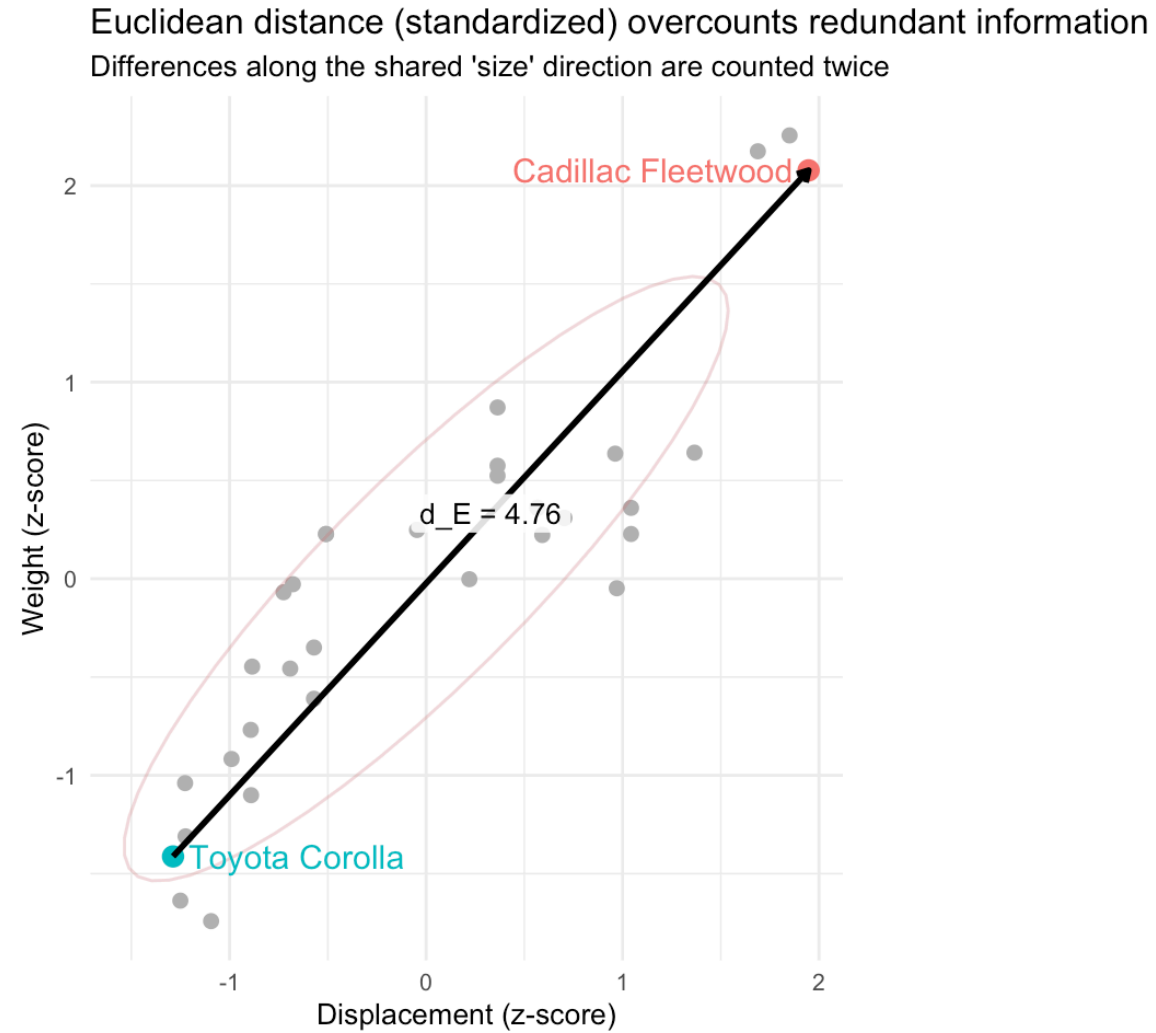
by variable differences: **independence-based**

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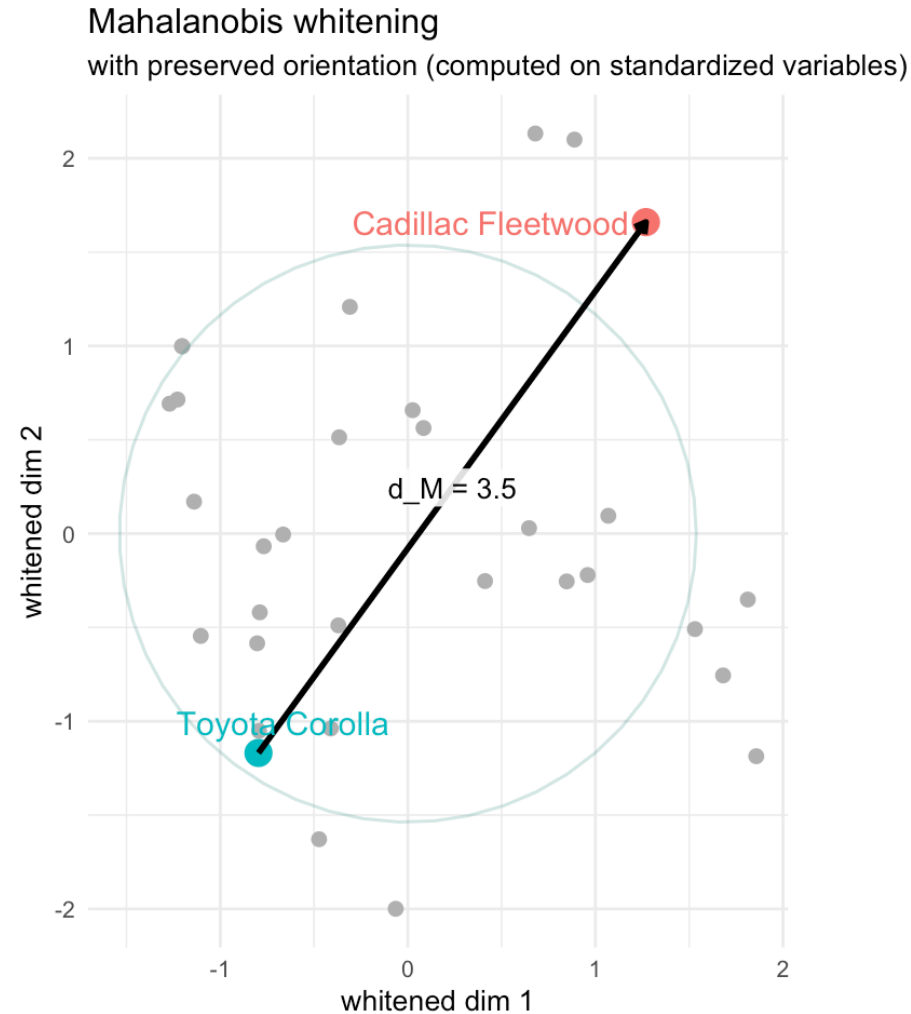
by variable differences: independence-based

The Euclidean distance $\rightarrow$ shared information is over-counted



accounting for inter-variable relations: association-based

The **Mahalanobis distance** → shared information is not over-counted



this is an **association-based** distance for continuous data

association-based distance

Association-based for continuous: **Mahalanobis distance**

Let $\mathbf{X}_n$ be $I \times Q_n$ a data matrix of I observations described by Q_n continuous variables, and let $\mathbf{S}$ the sample covariance matrix, the Mahalanobis distance matrix is

$$\mathbf{D}_{mah} = [\text{diag}(\mathbf{G}) \mathbf{1}^\top + \mathbf{1} \text{diag}(\mathbf{G})^\top - 2\mathbf{G}]^{\odot 1/2}$$

where

- $[\cdot]^{\odot 1/2}$ denotes the element-wise square root and $\mathbf{1}$ an I -dimensional vector of 1's.
- $\mathbf{G} = (\mathbf{C}\mathbf{X}_I)\mathbf{S}^{-1}(\mathbf{C}\mathbf{X}_I)^\top$ is the Mahalanobis Gram matrix
- $\mathbf{C} = \mathbf{I} - \frac{1}{I}\mathbf{1}\mathbf{1}^\top$ is the centering operator, $\mathbf{I}$ is an identity matrix of size I

association-based distance

Association-based for categorical: **total variation distance (TVD)**¹

The distance matrix $\mathbf{D}_{tvd}$ can be defined via the **delta framework**² upon properly defining the block-diagonal matrix $\mathbf{\Delta}$

Let $\mathbf{X}_{cat}$ be $I \times Q_c$ a data matrix of n observations described by Q_c categorical variables.

$$\mathbf{D} = \mathbf{Z}\mathbf{\Delta}\mathbf{Z}^T = [\mathbf{Z}_1 \quad \dots \quad \mathbf{Z}_{Q_c}] \begin{bmatrix} \mathbf{\Delta}_1 & & \\ & \ddots & \\ & & \mathbf{\Delta}_{Q_c} \end{bmatrix} \begin{bmatrix} \mathbf{Z}_1^T \\ \vdots \\ \mathbf{Z}_{Q_c}^T \end{bmatrix}$$

- in the framework, setting $\mathbf{\Delta}_j$ determines the categorical distance measure of choice (independent- or association-based)

association-based distance

Association-based for categorical: **total variation distance (TVD)**^{1 (2)}

Consider the empirical **joint probability** distributions stored in the off-diagonal blocks of $\mathbf{P}$:

$$\mathbf{P} = \frac{1}{n} \begin{bmatrix} \mathbf{Z}_1^\top \mathbf{Z}_1 & \mathbf{Z}_1^\top \mathbf{Z}_2 & \cdots & \mathbf{Z}_1^\top \mathbf{Z}_{Q_c} \\ \vdots & \ddots & \vdots & \vdots \\ \mathbf{Z}_{Q_c}^\top \mathbf{Z}_1 & \mathbf{Z}_{Q_c}^\top \mathbf{Z}_2 & \cdots & \mathbf{Z}_{Q_c}^\top \mathbf{Z}_{Q_c} \end{bmatrix}.$$

The block matrix $\mathbf{R}$ refer to the **conditional probability** distributions for each variable j given each variable i ($i, j = 1, \dots, Q_c, i \neq j$), stored in the block matrix

$$\mathbf{R} = \mathbf{P}_z^{-1}(\mathbf{P} - \mathbf{P}_z).$$

where $\mathbf{P}_z = \mathbf{P} \odot \mathbf{I}_{Q^*}$, and $\mathbf{I}_{Q^*}$ is the $Q^* \times Q^*$ identity matrix.

association-based distance

Association-based for categorical: **total variation distance (TVD)**¹ (3)

Let $\mathbf{r}_a^{ji}$ and $\mathbf{r}_b^{ji}$ be the rows of $\mathbf{R}_{ji}$, the (j, i) th off-diagonal block of $\mathbf{R}$.

The category dissimilarity between a and b for variable j based on the total variation distance (TVD) is defined as

$$\delta_{tvd}^j(a, b) = \sum_{i \neq j}^{Q_c} w_{ji} \Phi^{ji}(\mathbf{r}_a^{ji}, \mathbf{r}_b^{ji}) = \sum_{i \neq j}^{Q_c} w_{ji} \left[\frac{1}{2} \sum_{\ell=1}^{q_i} |\mathbf{r}_{a\ell}^{ji} - \mathbf{r}_{b\ell}^{ji}| \right],$$

where $w_{ji} = 1/(Q_c - 1)$ for equal weighting (can be user-defined).

TVD-based dissimilarity matrix is, therefore,

$$\mathbf{D}_{tvd} = \mathbf{Z} \Delta^{(tvd)} \mathbf{Z}^T.$$

association-based distance: a small example

Data

Consider two categorical variables:

- X_1 with categories A, B, C
- X_2 with categories u, v

```
# A tibble: 10 × 3  
  id X1  X2  
  <int> <fct> <fct>  
1     1  A    u  
2     2  A    u  
3     3  A    v  
4     4  B    u  
5     5  B    u  
6     6  B    v  
7     7  C    u  
8     8  C    v  
9     9  C    v  
10    10  C    v
```

Indicator matrices

$$\mathbf{Z}_1 = \begin{pmatrix} 1 & 0 & 0 \\ 1 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 1 \\ 0 & 0 & 1 \\ 0 & 0 & 1 \end{pmatrix}, \quad \mathbf{Z}_2 = \begin{pmatrix} 1 & 0 \\ 1 & 0 \\ 0 & 1 \\ 1 & 0 \\ 1 & 0 \\ 0 & 1 \\ 1 & 0 \\ 0 & 1 \\ 0 & 1 \\ 0 & 1 \end{pmatrix}.$$

association-based distance: from $\mathbf{Z}$ to $\mathbf{P}$

Let

$$\mathbf{Z} = [\mathbf{Z}_1, \mathbf{Z}_2].$$

The empirical co-occurrence matrix is

$$\mathbf{P} = \frac{1}{10} \mathbf{Z}^\top \mathbf{Z}.$$

For this example,

$$\mathbf{P} = \begin{pmatrix} 0.30 & 0 & 0 & 0.20 & 0.10 \\ 0 & 0.30 & 0 & 0.20 & 0.10 \\ 0 & 0 & 0.40 & 0.10 & 0.30 \\ 0.20 & 0.20 & 0.10 & 0.50 & 0 \\ 0.10 & 0.10 & 0.30 & 0 & 0.50 \end{pmatrix}.$$

diagonal blocks contain marginal information; off-diagonal blocks contain joint proportions

association-based distance: from $\mathbf{P}$ to $\mathbf{R}$

The diagonal part of $\mathbf{P}$ is

$$\mathbf{P}_z = \mathbf{P} \odot \mathbf{I}_{Q^*} = \text{diag}(0.30, 0.30, 0.40, 0.50, 0.50).$$

The block matrix of conditional profiles is

$$\mathbf{R} = \mathbf{P}_z^{-1}(\mathbf{P} - \mathbf{P}_z).$$

For this example,

$$\mathbf{R} = \begin{pmatrix} 0 & 0 & 0 & 0.67 & 0.33 \\ 0 & 0 & 0 & 0.67 & 0.33 \\ 0 & 0 & 0 & 0.25 & 0.75 \\ 0.40 & 0.40 & 0.20 & 0 & 0 \\ 0.20 & 0.20 & 0.60 & 0 & 0 \end{pmatrix}.$$

each off-diagonal block contains conditional profiles across variables

association-based distance: from Δ to $\mathbf{D}$

We collect the category dissimilarity matrices in a block-diagonal matrix:

$$\Delta^{(tvd)} = \begin{pmatrix} \Delta_1^{(tvd)} & 0 \\ 0 & \Delta_2^{(tvd)} \end{pmatrix}.$$

The observation-level categorical distance matrix is then

$$\mathbf{D}_{tvd} = \mathbf{Z} \Delta^{(tvd)} \mathbf{Z}^T = [\mathbf{Z}_1 \quad \mathbf{Z}_2] \begin{pmatrix} \Delta_1^{(tvd)} & 0 \\ 0 & \Delta_2^{(tvd)} \end{pmatrix} \begin{bmatrix} \mathbf{Z}_1^T \\ \mathbf{Z}_2^T \end{bmatrix}.$$

Equivalently,

$$\mathbf{D}_{tvd} = \mathbf{Z}_1 \Delta_1^{(tvd)} \mathbf{Z}_1^T + \mathbf{Z}_2 \Delta_2^{(tvd)} \mathbf{Z}_2^T.$$

category-level dissimilarities are translated into observation-level distances

Interaction-aware extensions

Cross-type interactions

Association-aware distances account for relations **within** variable blocks:

- continuous–continuous relations;
- categorical–categorical relations.

Cross-type structure

In mixed data, categorical differences may be meaningful because they are reflected in the continuous variables.

the next step is to make distances **interaction-aware**

How to measure interactions¹

Define Δ^{int} to account for continuous–categorical interactions and use it to augment Δ^{tvd} .

The mixed dissimilarity becomes

$$\mathbf{D}_{mix}^{(int)} = \mathbf{D}_{mah} + \mathbf{D}_{cat}^{(int)}.$$

where

$$\mathbf{D}_{cat}^{(int)} = \mathbf{Z}\tilde{\Delta}\mathbf{Z}^\top$$

and

$$\tilde{\Delta} = (1 - \alpha)\Delta^{tvd} + \alpha\Delta^{int}, \quad \alpha = \frac{1}{Q_c}.$$

What is Δ^{int} ?

The entry $\delta_{int}^j(a, b)$ measures how much the continuous variables help discriminate between observations choosing category a and those choosing category b for categorical variable j .

Category-pair classification problem

For each pair (a, b) :

- use the continuous variables as predictors;
- classify observations belonging to categories a and b ;
- use a nearest-neighbour rule in the continuous space.

Computing Δ_j^{int}

For each categorical variable j and each pair of categories (a, b) :

1. use $\mathbf{D}_{mah}$ to identify neighbours in the continuous space;
2. consider a proportion of neighbours, say $\hat{\pi}_{nn} = 0.1$;
3. classify observations using a prior-corrected decision rule;
4. compute balanced accuracy.

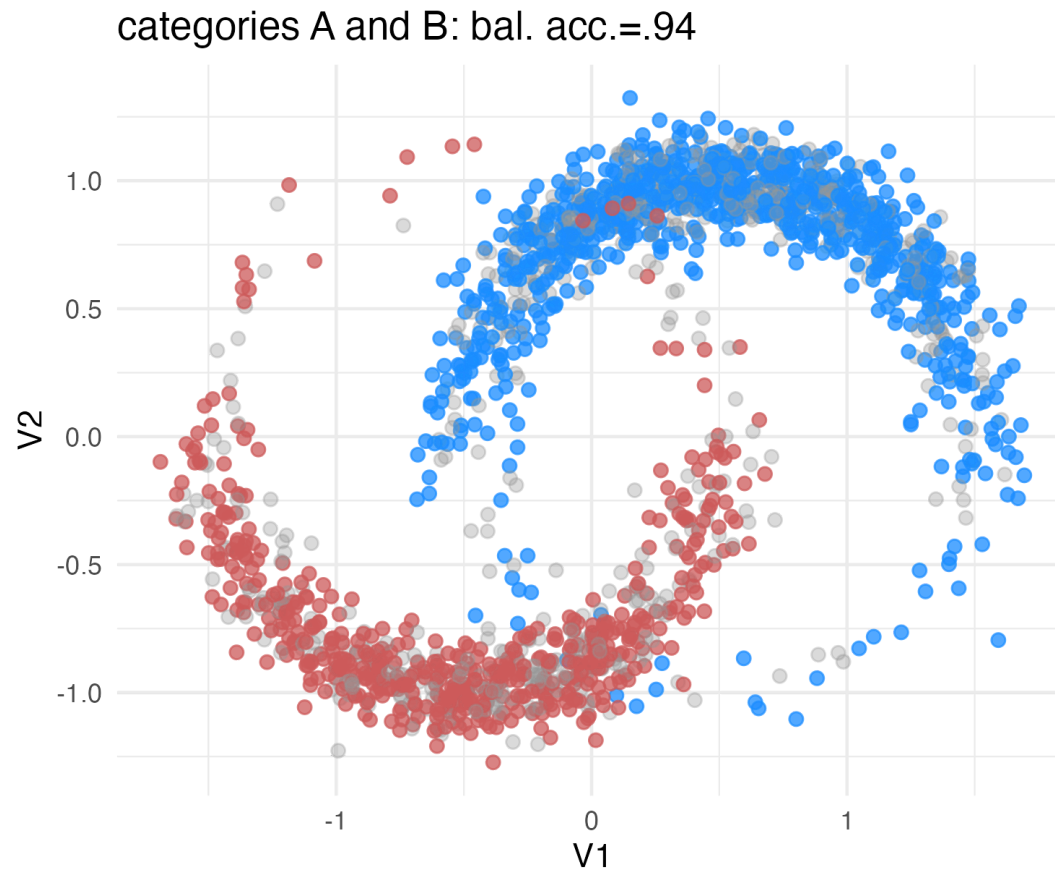
$$\delta_{int}^j(a, b) = \frac{1}{2} \left(\frac{\mathbf{true\ } a}{\mathbf{true\ } a + \mathbf{false\ } a} + \frac{\mathbf{true\ } b}{\mathbf{true\ } b + \mathbf{false\ } b} \right).$$

Computing Δ_j^{int}

For categorical variable j with q_j categories, compute $\frac{q_j(q_j-1)}{2}$ category-pair quantities.

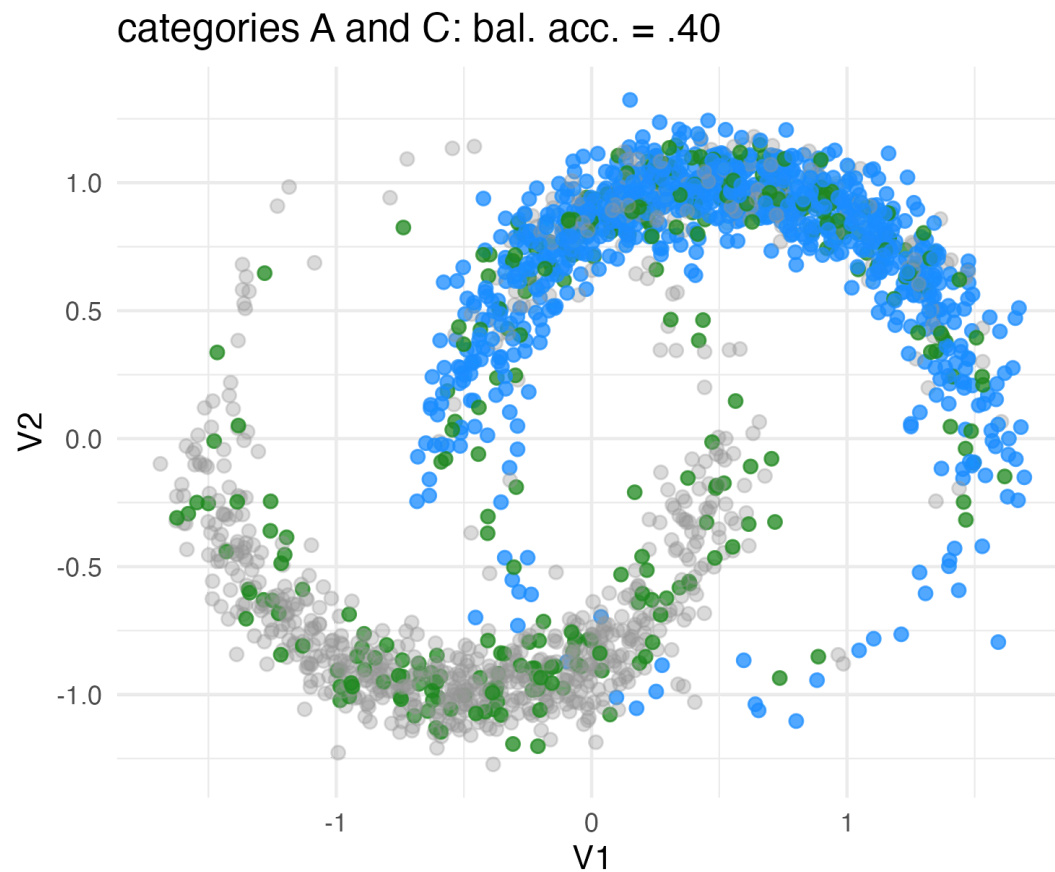
$$\Delta_{int} = \begin{pmatrix} 0 & \cdot & \cdot & \cdot \\ \cdot & 0 & \cdot & \cdot \\ \cdot & \cdot & 0 & \cdot \\ \cdot & \cdot & \cdot & 0 \end{pmatrix}$$

Computing Δ_j^{int}



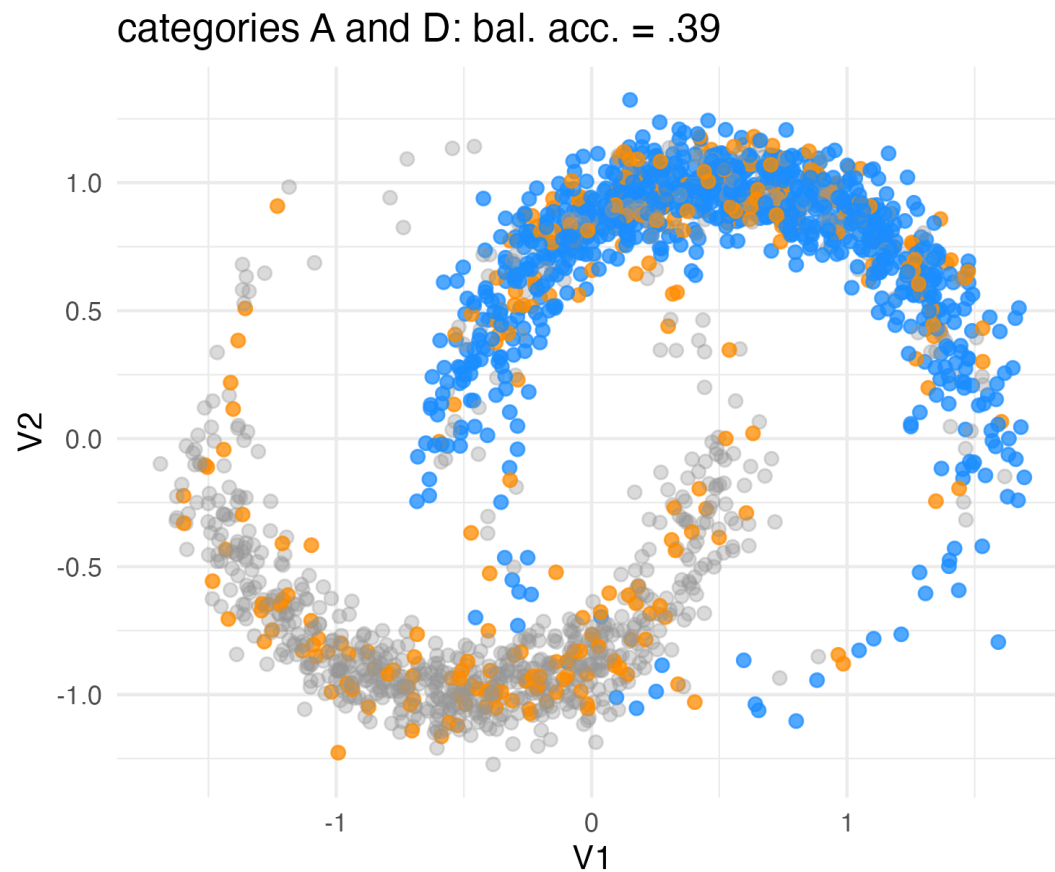
$$\Delta_{int} = \begin{pmatrix} 0 & 0.94 & \cdot & \cdot \\ 0.94 & 0 & \cdot & \cdot \\ \cdot & \cdot & 0 & \cdot \\ \cdot & \cdot & \cdot & 0 \end{pmatrix}$$

Computing Δ_j^{int}



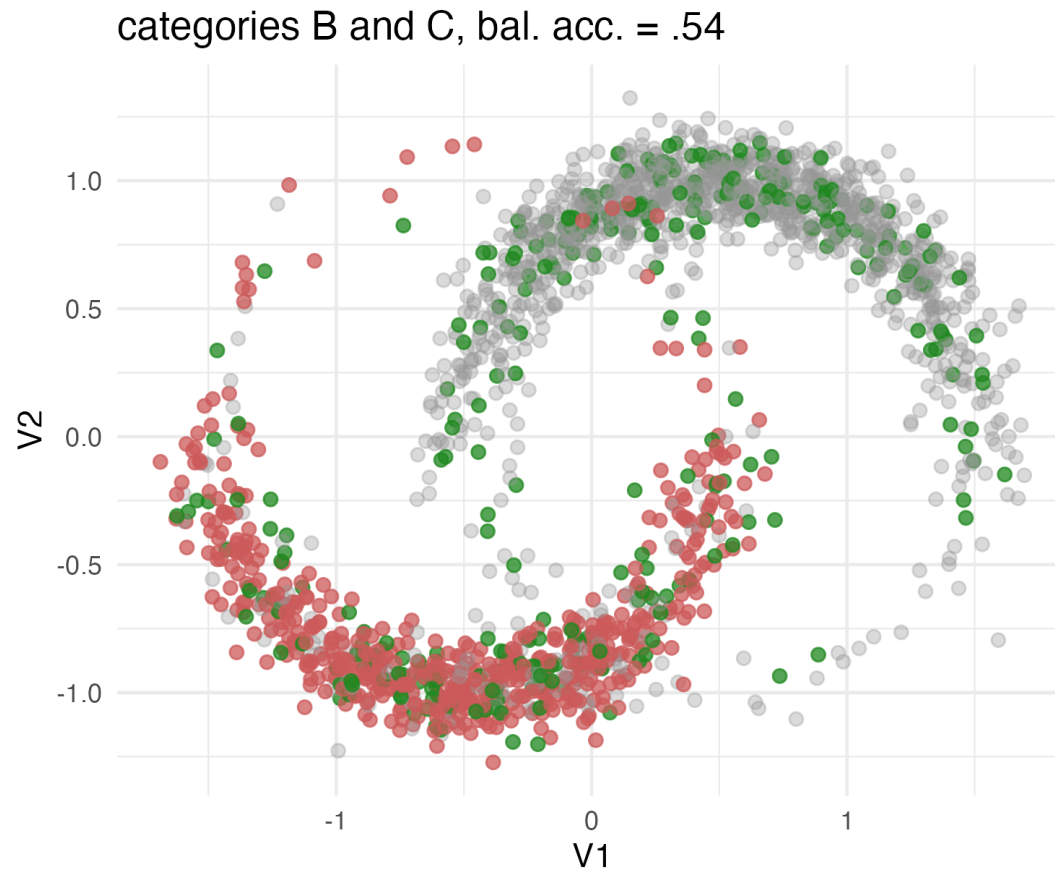
$$\Delta_{int} = \begin{pmatrix} 0 & 0.94 & \textcolor{red}{0.40} & \cdot \\ 0.94 & 0 & \cdot & \cdot \\ \textcolor{red}{0.40} & \cdot & 0 & \cdot \\ \cdot & \cdot & \cdot & 0 \end{pmatrix}$$

Computing Δ_j^{int}



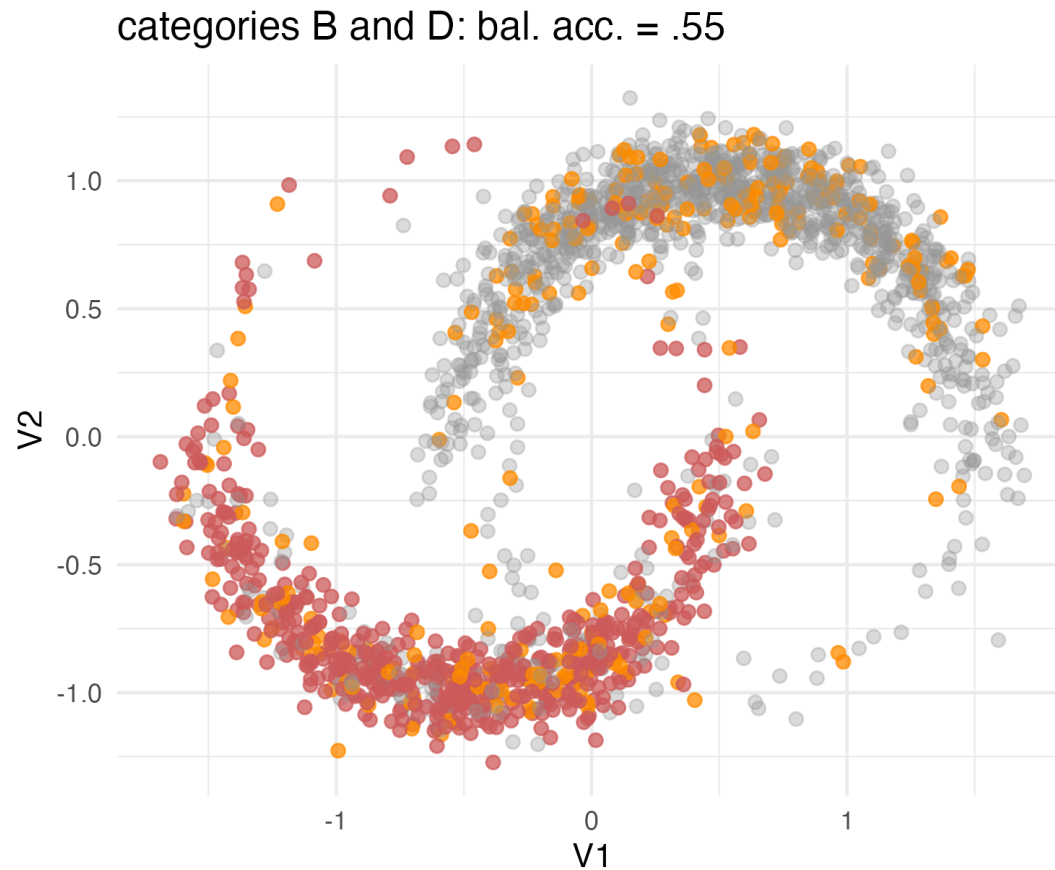
$$\Delta_{int} = \begin{pmatrix} 0 & 0.94 & 0.40 & \mathbf{0.39} \\ 0.94 & 0 & \cdot & \cdot \\ 0.40 & \cdot & 0 & \cdot \\ \mathbf{0.39} & \cdot & \cdot & 0 \end{pmatrix}$$

Computing Δ_j^{int}



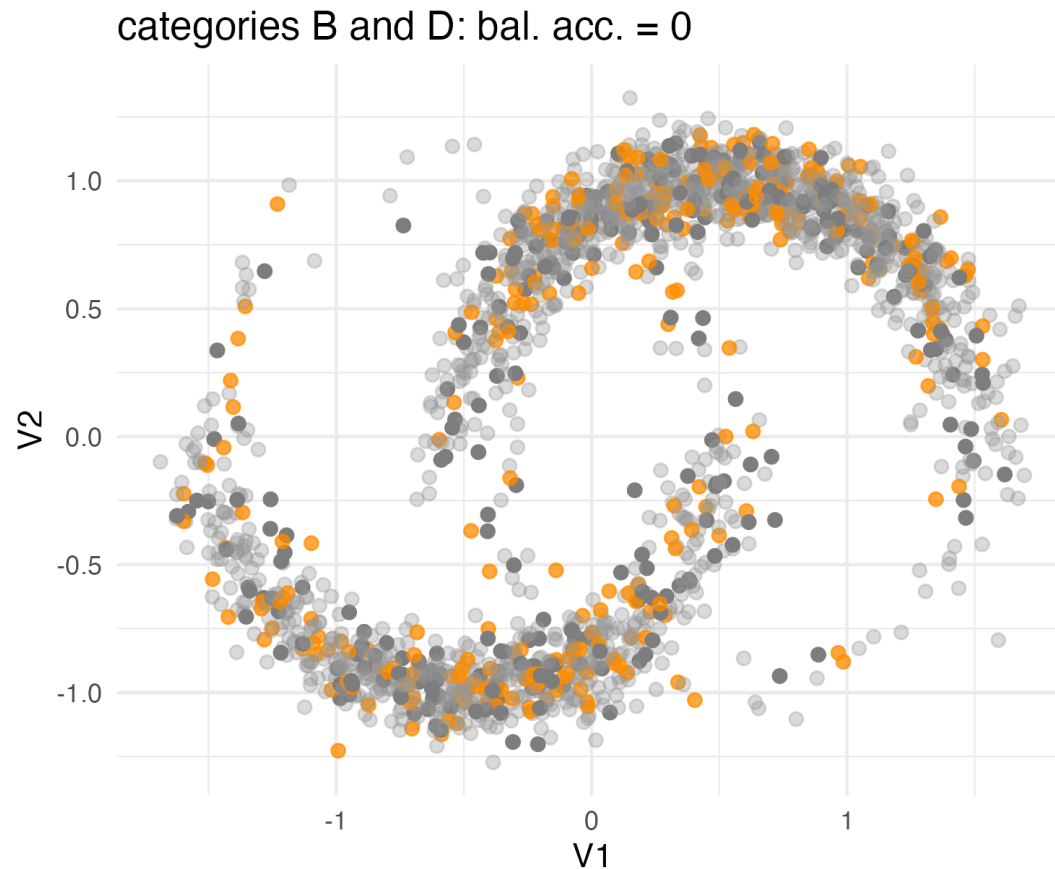
$$\Delta_{int} = \begin{pmatrix} 0 & 0.94 & 0.40 & 0.39 \\ 0.94 & 0 & \mathbf{0.54} & \cdot \\ 0.40 & \mathbf{0.54} & 0 & \cdot \\ 0.39 & \cdot & \cdot & 0 \end{pmatrix}$$

Computing Δ_j^{int}



$$\Delta_{int} = \begin{pmatrix} 0 & 0.94 & 0.40 & 0.39 \\ 0.94 & 0 & 0.54 & \mathbf{0.55} \\ 0.40 & 0.54 & 0 & . \\ 0.39 & \mathbf{0.55} & . & 0 \end{pmatrix}$$

Computing Δ_j^{int}



$$\Delta_{int} = \begin{pmatrix} 0 & 0.94 & 0.40 & 0.39 \\ 0.94 & 0 & 0.54 & 0.55 \\ 0.40 & 0.54 & 0 & 0 \\ 0.39 & 0.55 & 0 & 0 \end{pmatrix}$$

summarize category-pair separability in the continuous space

Just one-way interaction?

Result A — Factorization

Let

$$\mathbf{X} = [\mathbf{X}_{con} \ \mathbf{X}_{cat}]$$

contain n observations described by Q_d continuous and Q_c categorical variables, with

$$\mathbf{x}_i = (\mathbf{x}_{i_{con}}, \mathbf{x}_{i_{cat}}).$$

The joint distribution can be written as

$$f(\mathbf{x}_{i_{con}}, \mathbf{x}_{i_{cat}}) = f(\mathbf{x}_{i_{con}})f(\mathbf{x}_{i_{cat}} \mid \mathbf{x}_{i_{con}}).$$

This motivates using the continuous variables to quantify the separability of categorical configurations:

$$\mathbf{X}_{con} \longrightarrow \delta_{int}^j(a, b).$$

the asymmetry concerns the modelling direction, not the final pairwise dissimilarity

Just one-way interaction?

Result B — From a dissimilarity to a density

Starting from a non-negative dissimilarity from a cluster centre,

$$d(\mathbf{x}_i, \mathbf{c}_k) \geq 0, \quad d(\mathbf{c}_k, \mathbf{c}_k) = 0,$$

a probabilistic model can be constructed as (Ben-Israel & Iyigun, 2008; Hennig, Viroli, & Anderlucci, 2019)

$$f(\mathbf{x}_i; \mathbf{c}_k, \mathbf{S}_k) = g(\mathbf{c}_k, \mathbf{S}_k) \exp \{-\eta d(\mathbf{x}_i, \mathbf{c}_k)\}.$$

Its maximum is

$$M_k = f(\mathbf{c}_k; \mathbf{c}_k, \mathbf{S}_k) = g(\mathbf{c}_k, \mathbf{S}_k).$$

Therefore,

$$\log \frac{M_k}{f(\mathbf{x}_i; \mathbf{c}_k, \mathbf{S}_k)} = \eta d(\mathbf{x}_i, \mathbf{c}_k)$$

the reference-normalized log-density recovers the original dissimilarity

Just one-way interaction?

Result C — The continuous component

For two continuous vectors, define

$$d_{\text{mah}}(\mathbf{x}_{i_{\text{con}}}, \mathbf{x}_{i'_{\text{con}}}) = [(\mathbf{x}_{i_{\text{con}}} - \mathbf{x}_{i'_{\text{con}}})^{\top} \mathbf{S}^{-1} (\mathbf{x}_{i_{\text{con}}} - \mathbf{x}_{i'_{\text{con}}})]^{1/2}.$$

Applying Result B with

$$f_{\text{con}}(\mathbf{x}_{i_{\text{con}}} \mid \mathbf{x}_{i'_{\text{con}}}, \mathbf{S}) \propto \exp \{ -d_{\text{mah}}(\mathbf{x}_{i_{\text{con}}}, \mathbf{x}_{i'_{\text{con}}}) \}$$

gives

$$\log \frac{M_{\text{con}}}{f_{\text{con}}(\mathbf{x}_{i_{\text{con}}} \mid \mathbf{x}_{i'_{\text{con}}}, \mathbf{S})} = d_{\text{mah}}(\mathbf{x}_{i_{\text{con}}}, \mathbf{x}_{i'_{\text{con}}})$$

this recovers the ordinary Mahalanobis distance used in $\mathbf{D}_{mah}$

Just one-way interaction?

Result D — A categorical probability model

For categorical variable j , let $\mathcal{A}_j$ denote its category set and let

$$x_{ij_{cat}} = a, \quad x_{i'j_{cat}} = b.$$

Starting from a non-negative categorical dissimilarity $\delta^j(a, b)$, define

$$p_j(a \mid b) = \frac{\exp \{ -\delta^j(a, b) \}}{Z_j(b)},$$

where

$$Z_j(b) = \sum_{h \in \mathcal{A}_j} \exp \{ -\delta^j(h, b) \}.$$

this is a proper PMF and remains well defined when $\delta^j(a, b) = 0$

Just one-way interaction?

Result E — Recovering the categorical dissimilarity

Because

$$\delta^j(b, b) = 0,$$

the maximum of the categorical PMF is

$$M_j(b) = p_j(b \mid b) = \frac{1}{Z_j(b)}.$$

Therefore,

$$\begin{aligned} \log \frac{M_j(b)}{p_j(a \mid b)} &= \log \frac{Z_j(b)^{-1}}{Z_j(b)^{-1} \exp\{-\delta^j(a, b)\}} \\ &= \boxed{\delta^j(a, b)}. \end{aligned}$$

the normalizing constant cancels and the original dissimilarity is recovered exactly

Just one-way interaction? — Wrap-up

Apply Result E using the interaction-aware dissimilarity

$$\tilde{\delta}^j(a, b) = (1 - \alpha)\delta_{tvd}^j(a, b) + \alpha\delta_{int}^j(a, b).$$

For the whole categorical vector,

$$d_{\text{cat}}^{(int)}(\mathbf{x}_{i_{\text{cat}}}, \mathbf{x}_{i'_{\text{cat}}}) = \sum_{j=1}^{Q_c} \tilde{\delta}^j(x_{ij_{\text{cat}}}, x_{i'j_{\text{cat}}}).$$

Combining the two contributions gives

$$d(\mathbf{x}_i, \mathbf{x}_{i'}) = d_{\text{mah}}(\mathbf{x}_{i_{\text{con}}}, \mathbf{x}_{i'_{\text{con}}}) + d_{\text{cat}}^{(int)}(\mathbf{x}_{i_{\text{cat}}}, \mathbf{x}_{i'_{\text{cat}}})$$

and hence

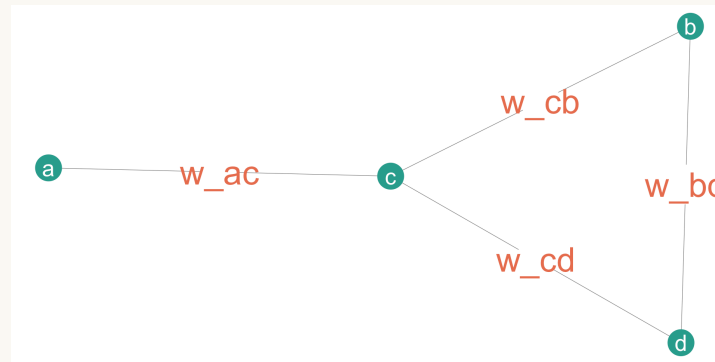
$$\mathbf{D}_{\text{mix}}^{(int)} = \mathbf{D}_{\text{mah}} + \mathbf{Z}\tilde{\Delta}\mathbf{Z}^\top.$$

the kNN computation of $\delta_{int}^j(a, b)$ remains unchanged

Spectral clustering: a graph partitioning problem

Graph representation

A graph representation of the data matrix $\mathbf{X}$: the aim is to cut it into K groups, or clusters.



The affinity matrix $\mathbf{A}$

The elements $\mathbf{w}_{ij}$ of $\mathbf{A}$ are high when observations i and j are likely to belong to the same group, and low otherwise.

.	a	b	c	d
a	0	0	w_{ac}	0
b	0	0	w_{cb}	w_{bd}
c	w_{ca}	w_{cb}	0	w_{cd}
d	0	w_{db}	w_{dc}	0

Spectral clustering: making the graph easy to cut

An approximate solution to the graph partitioning problem:

From distances to affinities

Start from the pairwise distance matrix $\mathbf{D}$ and build the affinity matrix

$$\mathbf{A} = \exp \left(-\frac{\mathbf{D}^2}{2\sigma^2} \right), \quad a_{ii} = 0.$$

The parameter σ controls the neighbourhood scale.

Normalized graph Laplacian

The normalized affinity matrix is

$$\mathbf{L} = \mathbf{D}_r^{-1/2} \mathbf{A} \mathbf{D}_r^{-1/2} = \mathbf{Q} \mathbf{\Lambda} \mathbf{Q}^T,$$

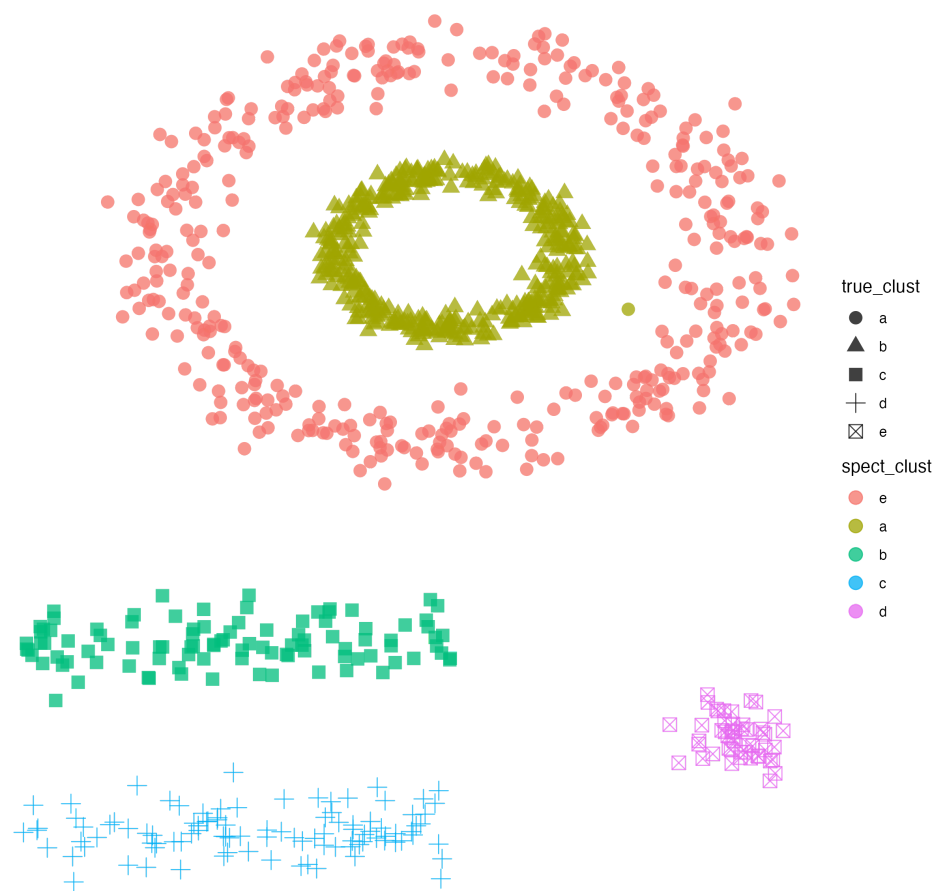
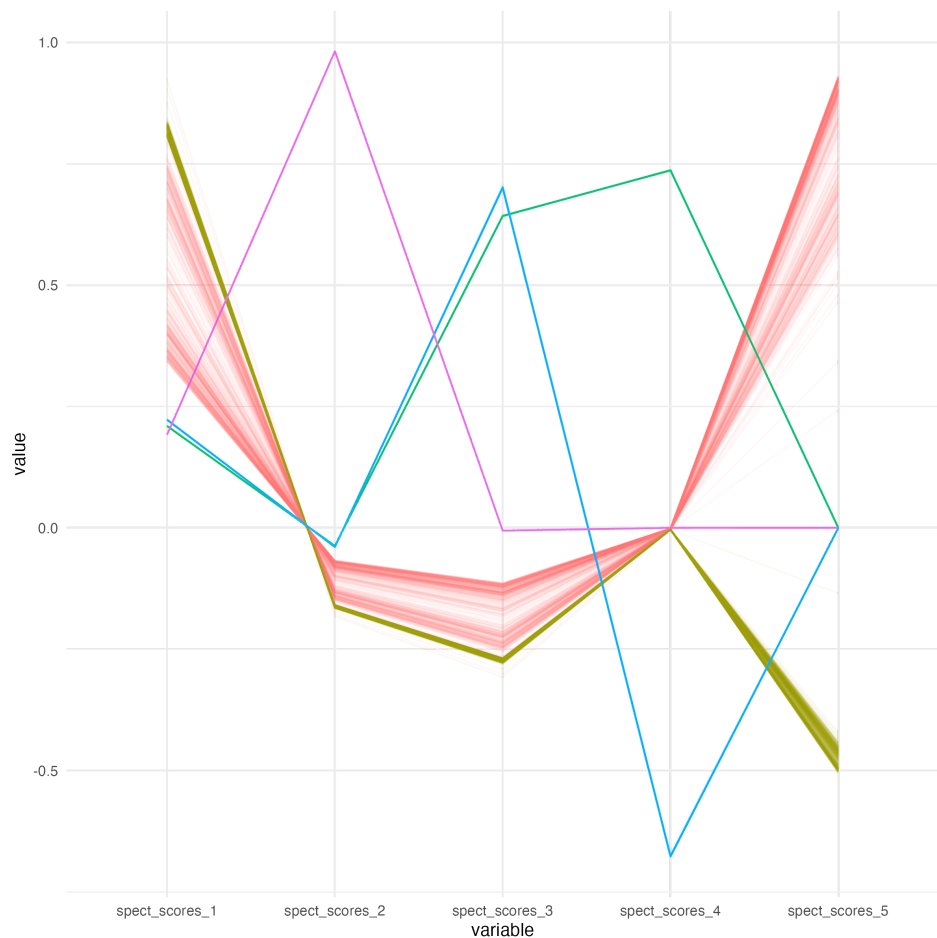
where $\mathbf{D}_r = \text{diag}(\mathbf{r})$, $\mathbf{r} = \mathbf{A} \mathbf{1}$, $\mathbf{1}$ is an n -dimensional vector of ones.

Spectral embedding

The spectral clustering solution is obtained by applying K -means to the rows of $\tilde{\mathbf{Q}}$, the matrix containing the first K eigenvectors of $\mathbf{L}$.

Why spectral clustering here?

Interaction-aware distances can encode local connectivity and non-convex structure.



Synthetic experiments

Simulation 1: associated mixed variables

General design

- two sample sizes: $I = 1000$ and $I = 2000$
- number of clusters fixed to $K = 3$
- 50 datasets generated for each configuration
- spectral clustering applied to each dissimilarity matrix
- performance measured by adjusted Rand index

Variable composition

Four scenarios were considered:

1. $Q_n = 15, Q_c = 15$
2. $Q_n = 10, Q_c = 20$
3. $Q_n = 20, Q_c = 10$
4. $Q_n = 15, Q_c = 15$
plus 5 numerical noise variables

Aim

The simulation creates mixed data with:

- correlated numerical variables;
- associated categorical variables;
- categorical variables depending on numerical variables;
- an additional noise scenario.

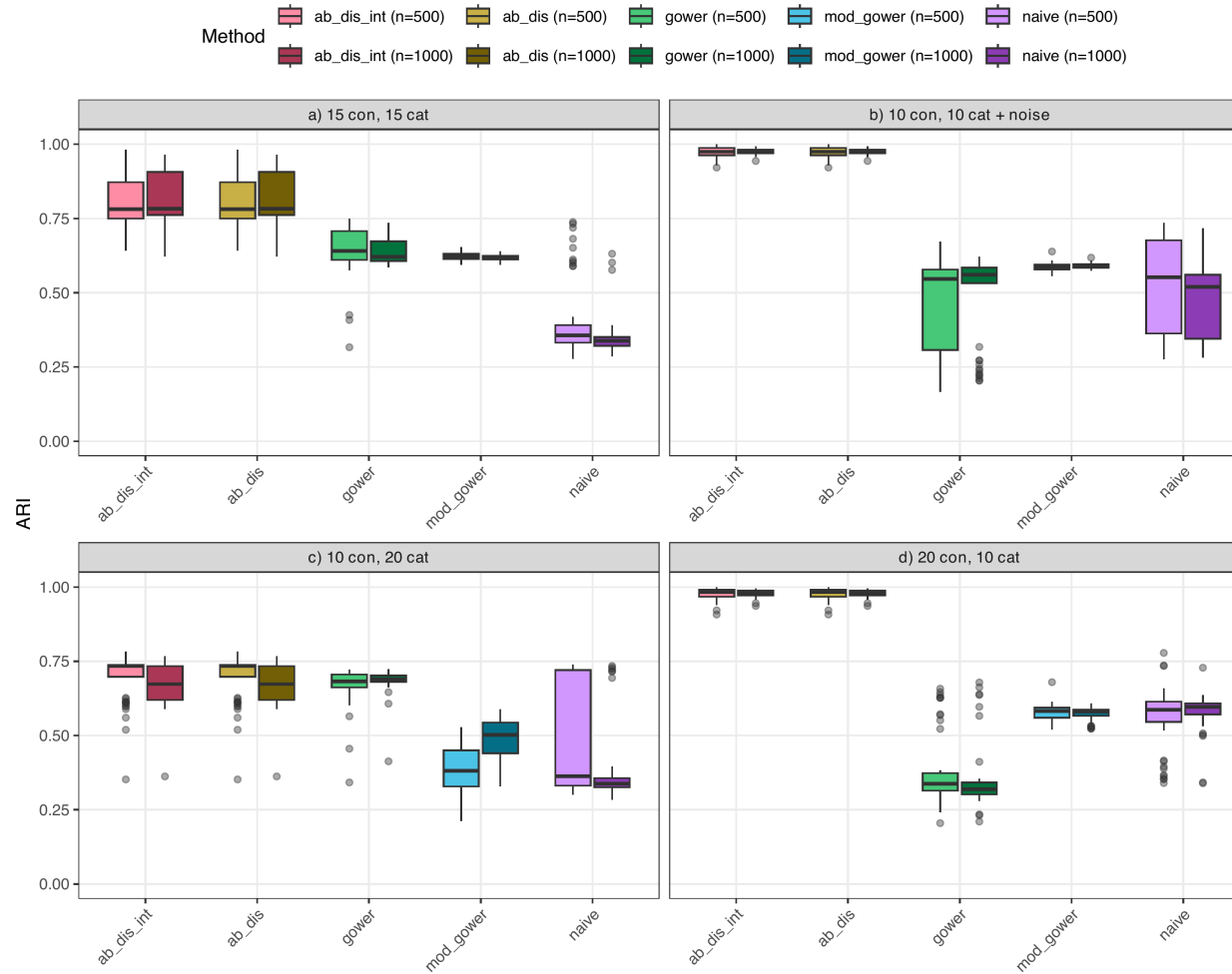
Simulation 1: associated mixed variables

Data-generating idea

1. Generate numerical variables from cluster-specific multivariate normal distributions.
 - different cluster means;
 - different correlation structures.
2. Generate binary categorical variables conditionally on the numerical variables.
 - the first categorical variable depends on the numerical block;
 - later categorical variables depend on the numerical block and on previous categorical variables.

the design induces numerical correlations, categorical associations, and numerical–categorical dependence

Simulation 1: results



- association-based distances are generally the most accurate;
- when continuous variables dominate, AB distances achieve almost perfect recovery;
- standard mixed-data distances are more sensitive to variable composition and noise.

Simulation 2: interaction-driven clusters

Design

- $I = 500, 1000$
- six continuous variables and three categorical variables
- V_4, V_5, V_6 generated independently from $N(0, 1)$
- V_1, V_2, V_3 generated conditionally on C_1 and C_2

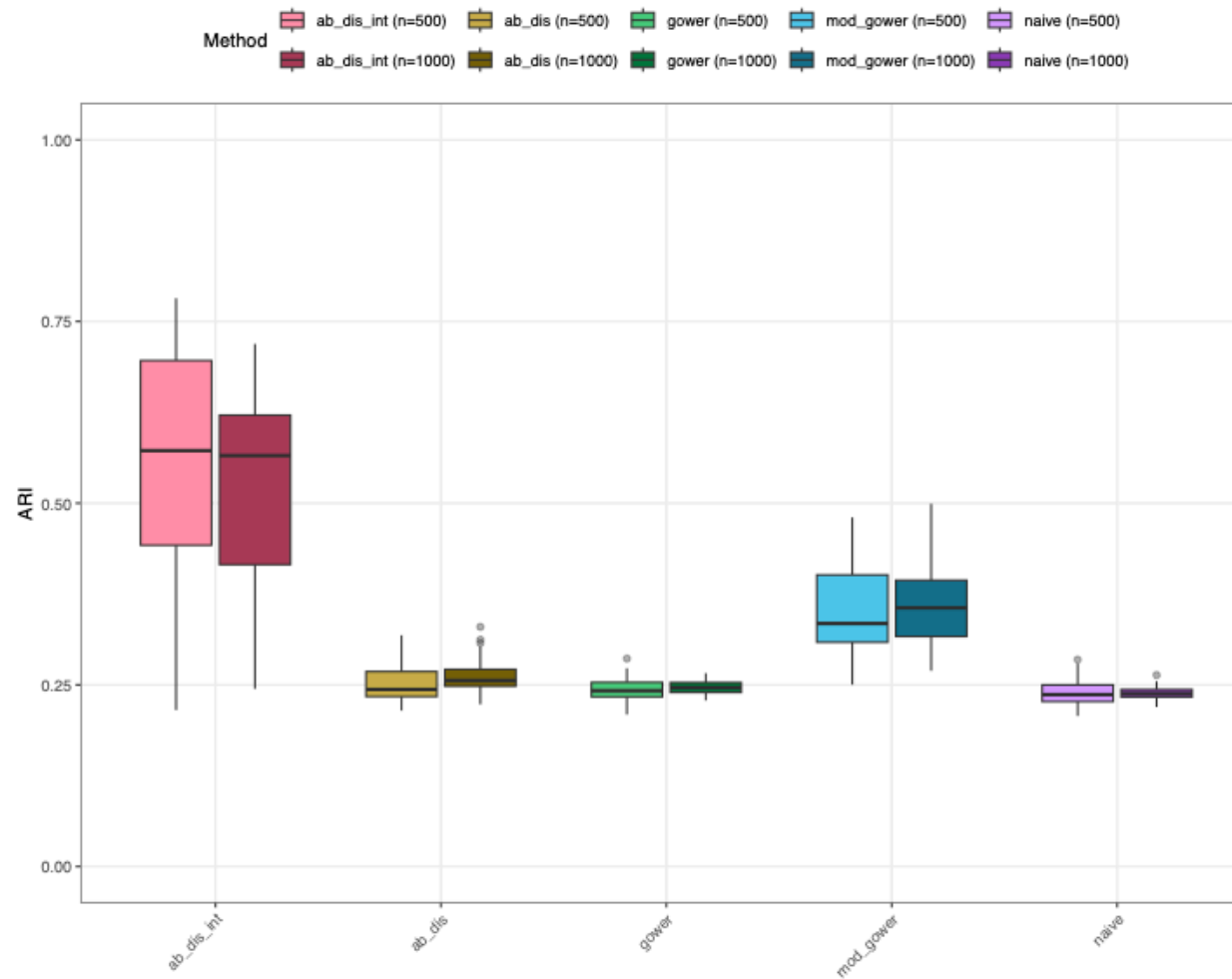
Main feature

The clusters are not defined by continuous variables alone or categorical variables alone.

They are defined by their **cross-type interaction**.

this is the setting where Δ^{int} should matter

Simulation 2: results



- `ab_dis_int` clearly outperforms all competitors;
- distances without cross-type interaction remain close to chance-level separation;
- the result supports explicitly encoding continuous–categorical interactions.

Wrap-up

Preprocessing is modelling

Distance-based learning makes no exception.

Distance choices are contextual

Some choices are domain-driven; others depend on the data structure and the downstream task.

Towards a common ground

Similar distance-based ideas often appear under different names across statistics, machine learning, econometrics, psychometrics, and operational research.

A package ecosystem such as **manydist** can make these choices easier to compare and reuse.

Main references

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Backup slides

Simulation 1: data generation

Numerical variables

Numerical variables are generated from cluster-specific multivariate normal distributions:

$$\mathbf{X}_n \mid C_k \sim N(\boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k).$$

Cluster proportions:

$$P(C_1) = 0.20, \quad P(C_2) = 0.30, \quad P(C_3) = 0.50.$$

Cluster-specific structure:

- C_1 : means equal to **0**, correlations **0.5**;
- C_2 : means equal to **4**, correlations **0.8**;
- C_3 : means **−1** for odd variables and **5** for even variables, correlations **−0.5**.

All variances are set to **1**.

Simulation 1: data generation

Categorical variables

Categorical variables are binary and generated recursively.

For categorical variable X_j^c ,

$$X_j^c \mid C_k \sim \text{Bernoulli}(p_{jk}),$$

with

$$p_{jk} = \left[1 + \exp \left(\boldsymbol{\beta}_k^\top \tilde{\mathbf{X}}_j \right) \right]^{-1}.$$

For $j = 1$,

$$\tilde{\mathbf{X}}_j = (1, \mathbf{X}_n),$$

while for $j > 1$,

$$\tilde{\mathbf{X}}_j = (1, \mathbf{X}_n, X_1^c, \dots, X_{j-1}^c).$$

The Gaussian special case

Gaussian reference model

Assume

$$f(\mathbf{x}_{i_{con}} \mid \mathbf{x}_{i'_{con}}, \mathbf{S}) \\ = \frac{1}{(2\pi)^{Q_d/2}} |\mathbf{S}|^{-1/2} \exp \left\{ -\frac{1}{2} (\mathbf{x}_{i_{con}} - \mathbf{x}_{i'_{con}}) \mathbf{S}^{-1} (\mathbf{x}_{i_{con}} - \mathbf{x}_{i'_{con}})^\top \right\}.$$

The maximum occurs at

$$\mathbf{x}_{i_{con}} = \mathbf{x}_{i'_{con}},$$

and hence

$$M = f(\mathbf{x}_{i'_{con}} \mid \mathbf{x}_{i'_{con}}, \mathbf{S}) = \frac{1}{(2\pi)^{Q_d/2}} |\mathbf{S}|^{-1/2}.$$

The reference-normalized log-density is therefore

$$\begin{aligned} \log \frac{M}{f(\mathbf{x}_{i_{con}} \mid \mathbf{x}_{i'_{con}}, \mathbf{S})} &= \frac{1}{2} (\mathbf{x}_{i_{con}} - \mathbf{x}_{i'_{con}}) \mathbf{S}^{-1} (\mathbf{x}_{i_{con}} - \mathbf{x}_{i'_{con}})^\top \\ &= \boxed{\frac{1}{2} d_{\text{mah}}^2(\mathbf{x}_{i_{con}}, \mathbf{x}_{i'_{con}})}. \end{aligned}$$

Gaussian model: $\frac{1}{2} d_{\text{mah}}^2$

model used in the main derivation: d_{mah}